

STOWLINE V1.0 · PRD & SRS SUMMARY

Stowline System Logic and Data Model

A 3D container stowage planner: ERD, business rules, rule engine, stability, workflow and API

MV Nusantara Pioneer · 8,500 TEU

Voyage 042W

Port call Singapore (SGSIN)

CONTEXT

Product and User Roles

A vessel planner places containers on a ship in 3D, and every rule violation is caught at the moment of placement, not after the plan is finished.

1,240

containers on the load list

312

already planned

2,740

containers on board

6

ports in the rotation

Role	Can do in Stowline
Vessel planner	Create and edit plans, place containers, validate, save, send for review
Senior planner	Open any plan, approve it, or return it to Draft with a comment
Terminal planner	View and export approved plans
Chief officer	View stability and violations, read only

The demo has no login: a role switcher in the account menu changes what the user can do.

BUSINESS PROCESS

12-Step Process Flow

Vessel planner

Stowline

Senior planner

Terminal planner

1 Open the port call

Voyages listed with ETD, progress, violations, status

2 Load the arrival condition

Containers already on board from earlier ports

3 Import the load list

This port's containers are added, each row checked

4 Place containers

Valid, warning and invalid slots marked before the drop

5 Check the rules

Re-checked on every move, violations listed

6 Fix violations

Show moves to the problem, Apply fix makes the move

7 Check stability

GM, trim, list, drafts, BM and SF against limits

8 Play the port rotation

What leaves at each port and the restows needed

9 Save & send for review

New version saved, status set to In review

10 Review

Approve, or return to Draft with a comment

11 Hand over

Approved plan exported; the plan is locked

12 Sail

Departure condition becomes the arrival condition at the next port

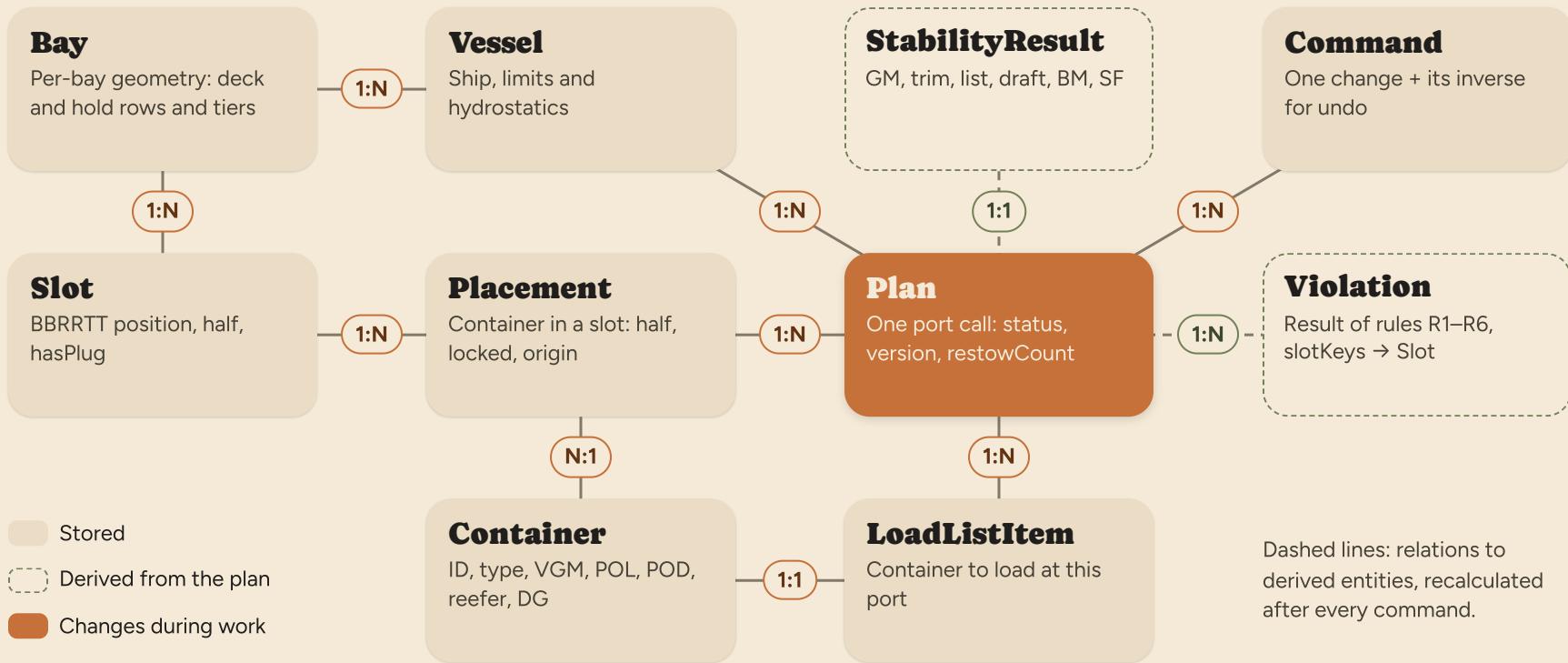
01

Model Data

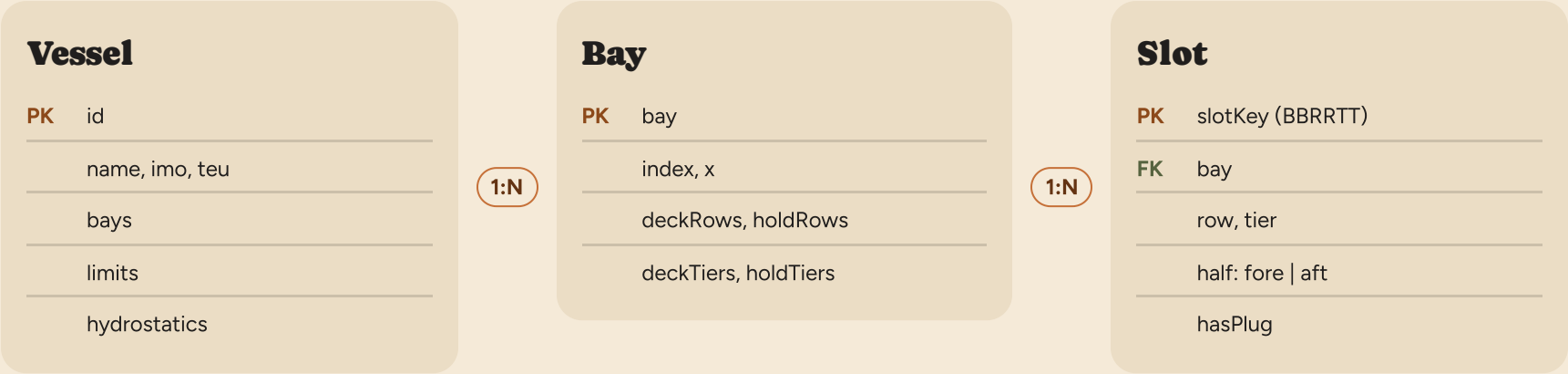
Ten domain entities. A plan is the only one that changes while the planner works.

DATA MODEL

ERD Overview: 10 Entities



ERD Detail: Vessel and Slot Geometry

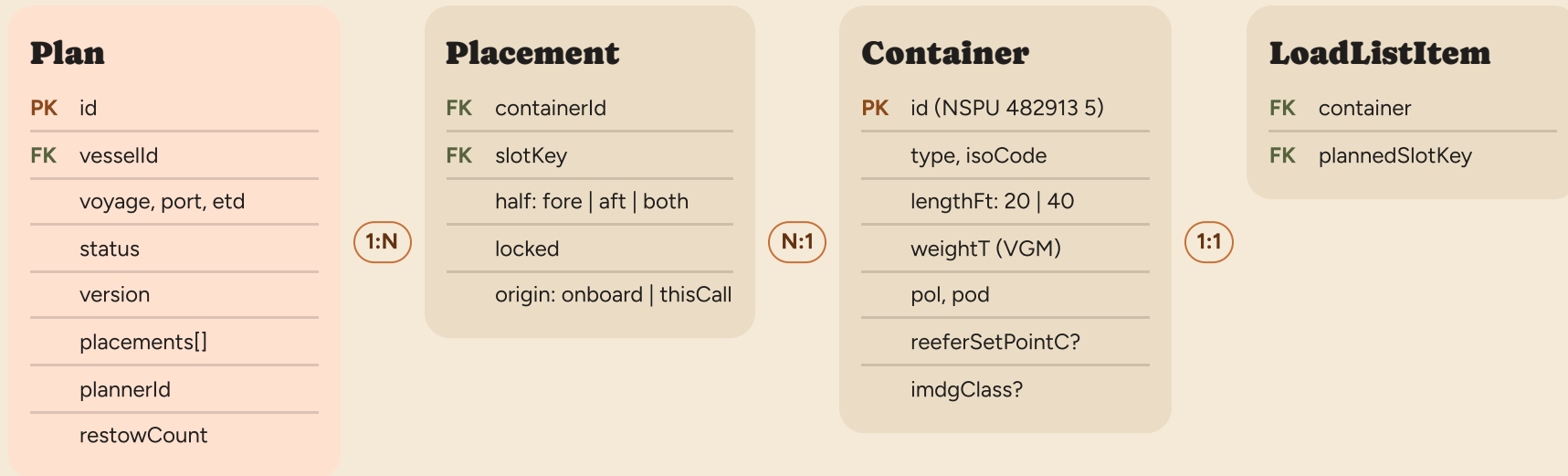


22 forty-foot bays, numbered 02 to 86 in steps of 4, with 4,292 forty-foot slots.

Deck tiers 82 to 92, hold tiers 02 to 16. Forward bays are narrower and shallower.

A slot exists only where the bay geometry has it. 404 slots have a reefer plug.

ERD Detail: Plan, Placement and Cargo



Plan status: draft, in_review or approved. Plan 1:N LoadListItem for this port call.

Each 40ft slot has a fore and an aft half, so two 20ft containers fit. Placement N:1 Slot.

Types: 20GP, 40GP, 40HC, RF, TK, OT. plannedSlotKey is empty until the container is placed.

ERD Detail: Violations, History and Stability

Plan 1:N · derived

Violation

PK id (stack:18-4-D)

rule: stack | reefer | dg | overstay | twenty
| heavy

severity: error | warning

FK slotKeys[]

message

data: restows, overT, port

The id is stable for the same rule and slot, so the UI can tell new violations from resolved ones.

Plan 1:N · stored

Command

kind: place | move | unplace | swap | lock |
unlock

payload

inverse

History stores each command and its inverse, not a full copy of the plan.

Plan 1:1 · never stored

StabilityResult

gm, trim, list

draftFwd, draftAft

bmPct, sfPct

bmCurve, sfCurve

Recalculated from the whole plan after every command, so undo and redo never drift.

02

System Logic

17 business rules, 6 engine rules, the placement check, stability, commands, plan status, versions and the API.

Slot, Stack and Cargo Rules (BR-01–BR-09)

ID	Rule	Result if broken
BR-01	A slot is addressed as bay, row, tier (BBRRTT). 20ft uses odd bays, 40ft even bays	Slot does not exist
BR-02	A container goes only into an empty slot with a container directly below, or the lowest tier	Drop refused
BR-03	Only the top container of a stack can be picked up, moved or unplaced	Action disabled
BR-04	A locked container cannot be moved, unplaced or swapped until unlocked	Action disabled
BR-05	Stack weight stays within the limit: 90.0 t on deck and 210.0 t in the hold	Error
BR-06	A reefer sits only in a slot with a power plug	Error
BR-07	Incompatible dangerous goods classes do not sit next to each other	Error
BR-08	A container for an earlier port does not sit under one for a later port	Error + restow count
BR-09	No 20ft on a 40ft. A 40ft sits on 20ft only when both halves below are filled	Error

Drop, Stability and Plan Rules (BR-10–BR-17)

ID	Rule	Result if broken
BR-10	A container is not more than 10.0 t heavier than the one below it	Warning
BR-11	A drop that causes an error is refused. A drop that causes only a warning is allowed	Container returns to origin
BR-12	Stability within limits: $GM \geq 1.20$ m, $trim \leq 1.50$ m, $list < 2.0^\circ$, BM and $SF \leq 100\%$	Gauge shows “Limit”
BR-13	Caution before a limit: $GM < 1.40$ m, $trim > 1.00$ m, $list > 0.3^\circ$, BM or $SF > 85\%$	Gauge shows “Check”
BR-14	A plan is Draft, In review or Approved. It cannot be approved while any error remains; warnings do not block	Approve disabled
BR-15	An approved plan is locked. A change after approval starts a new Draft version	Edit refused
BR-16	Every save has a version number. A save based on an old version is refused; the planner's moves are kept	Conflict message
BR-17	A container loaded at an earlier port can be moved here, but the move counts as a restow	Port call restow count rises

RULE ENGINE

Six Rules R1–R6

Error

R1 Stack weight

Stack weights add up to more than 90.0 t (deck) or 210.0 t (hold)

Fix: move the top container to the nearest valid slot, same bay first

Error

R2 Reefer power

A reefer sits in a slot without a plug

Fix: move it to the nearest valid slot with a plug

Error

R3 DG segregation

Incompatible classes are neighbours: left, right, above, below, or same row and tier in the next bay

Fix: move one to a valid deck slot at least two bays away

Error

R4 Overstow

Containers above it go to a later port. Restows = the number of those containers

Fix: reorder the stack so later ports sit lower, as one command

Error

R5 20ft and 40ft

A 20ft directly on a 40ft, or a 40ft on 20ft with one half below empty

Fix: move it to a valid slot on a 20ft stack or the lowest tier

Warning

R6 Heavy over light

A container is more than 10.0 t heavier than the one directly below

Fix: swap the two containers

Each rule is a pure function of the plan and the vessel. A fix is offered only when it creates no new error. Sort order: errors first, then R1 to R6.

DG Segregation Table and Golden Fixture

IMDG class	Must not sit next to
3 Flammable liquids	5.1, 2.1, 1.4
5.1 Oxidizing substances	3, 2.1
2.1 Flammable gases	3, 5.1
1.4 Explosives	3

Demonstration data, not the full IMDG segregation table.

Rule	Slot	Detail
R1 Stack weight	180488	Stack 18-04 deck at 96.4 t of 90.0 t
R2 Reefer power	220610	NSPU 220417 3 has no plug
R3 DG	140284, 140484	IMDG 3 next to IMDG 5.1
R4 Overstow	100382	Colombo under Rotterdam, 2 restows
R4 Overstow	420882	Jebel Ali under Hamburg, 1 restow
R5 20ft/40ft	290284	NSPU 318204 6 on a 40ft stack in bay 30
R6 Heavy/light	460612	30.2 t above 8.4 t

The seeded plan always gives 7 violations (6 errors, 1 warning). This is the main regression test.

PLACEMENT CHECK

Placement Check Order

1 The slot exists in the bay geometry Fail → not a target

2 The slot is empty, apart from the container being moved Fail → not a target

3 Lowest tier, or the slot below is filled Fail → not a target

4 R1–R5 tested on the stack after the drop **Fail → invalid, first reason shown**

5 R6 tested **Fail → valid with warning**

Stops at the first structural failure (steps 1–3).

Check result

Target: yes / no

Valid: yes / no

List of errors

List of warnings

New stack weight and its limit

An invalid drop is refused: the container returns in 220 ms, the slot shakes once and a message gives the reason.

Stability Calculation and Limits

For each container i: weight w , position x, y, z

$$\Delta = \Delta_o + \sum w_i$$

$$KG = (\Delta_o \cdot KG_o + \sum w_i z_i) / \Delta$$

$$GM = KM - KG$$

$$\text{trim} = \text{trim}_o + \sum w_i (LCF - x_i) / (100 \cdot \text{MTC})$$

$$\text{list} = \arctan((M_o + \sum w_i y_i) / (\Delta \cdot GM))$$

$$T_{\text{mean}} = T_{\text{ref}} + (\Delta - \Delta_{\text{ref}}) / (100 \cdot \text{TPC})$$

$$T_{\text{fwd}} = T_{\text{mean}} - \frac{1}{2} \text{trim} \cdot T_{\text{aft}} = T_{\text{mean}} + \frac{1}{2} \text{trim}$$

Recalculated from the whole plan after every command. Drag preview
= model(plan + candidate command) - model(current plan).

Value	Check	Limit
GM	< 1.40 m	< 1.20 m
Trim (either direction)	> 1.00 m	> 1.50 m
List (either side)	> 0.3°	≥ 2.0°
Bending moment, shear force	> 85%	> 100%

Seeded plan values

GM 1.84 m

Trim 0.62 m by the stern

List 0.4° to port

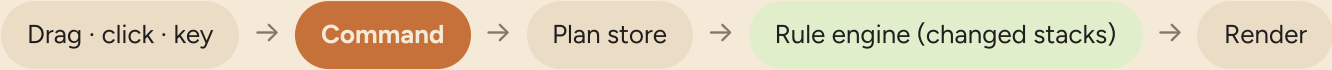
Draft 12.10 / 12.72 m

BM 78% · SF 64%

Strength curves: 61 stations. The weight change per bay is balanced, integrated for SF and BM, then divided by the limits (15,000 t, 550,000 t m).

COMMANDS & HISTORY

Commands, Undo and Redo



Command	Inverse (undo)
place (containerId → to)	unplace (to)
move (from → to)	move (to → from)
unplace (from)	place back at from
swap (a, b)	swap (b, a)
lock (at)	unlock (at)

Apply fix runs as one command and passes the same placement check.

Undo: Ctrl/Cmd + Z. Redo: Ctrl/Cmd + Shift + Z. Every command adds an activity log entry.

Property test: a command followed by its inverse returns the same plan.

Unsaved commands are kept in the browser and restored after a reload.

WORKFLOW

Plan Status and Save Versions



INTERFACE

Ten API Endpoints

Method	Path	Purpose	Error
GET	/api/plans	List port calls	
POST	/api/plans	Create a Draft plan	422 invalid input
GET	/api/plans/:id	Load one plan + version	404
PUT	/api/plans/:id	Save (baseVersion + command)	409 newer version, 422 rule
POST	/api/plans/:id/status	Change status + comment	403 role, 409 errors remain
GET	/api/plans/:id/load-list	Containers to load	404
POST	/api/plans/:id/load-list/import	Import rows from a file	422 not valid JSON
GET	/api/plans/:id/activity	Activity log	404
GET	/api/plans/:id/export	Download an approved plan	409 not approved
GET	/api/vessels/:id	Geometry, limits, hydrostatics	404

Version 1: Mock Service Worker answers with a 150–400 ms delay and plans are kept in IndexedDB. Every error has a code, message and details.